INTERNAL QUALITY ASSURANCE CELL (IQAC)

ST. JOSEPH'S COLLEGE (AUTONOMOUS)

Special Heritage Status Awarded by UGC | Accredited at A++ Grade (Cycle IV) by NAAC | Tiruchirappalli – 620 002

THEORY COURSE PLAN PORTAL • SYSTEM SPECIFICATION & ARCHITECTURE

Technical Stack, Institutional Workflows, OBE Mapping & Security Matrix (AY 2026–2027)

1. Executive Summary & Purpose

The **IQAC Theory Course Plan Portal** is an enterprise web application designed for St. Joseph's College (Autonomous), Tiruchirappalli. It streamlines the preparation, review, departmental approval, and institutional academic audit of theory course plans for **34 Academic Departments** (Shift I & Shift II) across all Even Semesters (Semesters II, IV, VI). The system enforces **Outcome-Based Education (OBE)** standards, Bloom's Taxonomy Cognitive Level mapping (K1 to K6), Continuous Internal Assessment (CIA) rules, JosTEL digital pedagogy, and seamless Head of Department (HOD) oversight.

2. Technology Stack & Architectural Specifications

The application employs a decoupled, robust full-stack architecture engineered for high availability, sub-second load latency, zero frontend bloat, and enterprise-grade data persistence.

Architectural Layer	Technologies & Frameworks	Key Capabilities & Implementation Details
Frontend UI / UX	Vanilla JavaScript (ES6+), Tailwind CSS (JIT Engine), Lucide Vector Icons	• Lightweight Single-Page Application (SPA) architecture with 0ms build compile requirement. • Responsive flexbox & CSS grid engine responsive from 360px mobile viewports to 4K displays. • Client-side reactive state store with dynamic DOM updates. • High-fidelity 1:1 print engine matching official college stationary formats.
Portal Core & Navigation	Next.js 14 (App Router), React 18 Server Components, TypeScript, PM2	• Serves the root institutional portal (`https://sjciqac.frontierwox.in/login`). • Automated process supervisor via PM2 (Process ID 0) running on Node.js 20 LTS. • Type-safe contracts and secure session handoffs.
Backend API Services	PHP 8.x REST API Engine, JSON Schema Engine	• RESTful micro-endpoints (`api/course_plan.php`) handling `save`, `get`, `delete`, `approve`, `reject`, and `list` operations. • Server-side payload sanitization protecting against XSS and parameter tampering. • Native JSON encoding/decoding with zero third-party latency.
Persistence & Data Layer	MySQL Enterprise Server, Dual-Store Fallback Engine	• ACID-compliant relational storage storing course plans, approvals, and faculty rosters. • High-availability dual-store engine with synchronous JSON datastore backup ensuring zero data loss during network disruptions.
Hosting & Infrastructure	Ubuntu Linux 24.04 LTS, Nginx Reverse Proxy, Let's Encrypt SSL/TLS	• Dedicated production cloud server (IP: 66.116.252.137) with Nginx HTTP/2 reverse proxy. • Dynamic URL rewrites (`/course-plan/login` → `/?login`). • Strict HTTPS encryption with automated SSL renewal. • Production Git versioning and automated asset cache-busting.

3. Functional Modules & Feature Specifications

The system is organized into modular functional components mapping to institutional teaching-learning requirements:

A. Course Information & OBE Curriculum Auto-Fetch (Part-I)

- **Personalized Faculty Greeting:** Dynamically greets faculty upon selecting their name from the verified departmental roster (`Welcome <>`) and surfaces tailored submission instructions.
- **Automated Syllabus Ingestion:** Comprehensive repository indexing 2023 & 2025 UG/PG regulation curricula across 34 departments. Automatically populates Course Title, Course Category (CC, DS, ES, GE, AE, VE), Total Hours, and Weekly Lecture Hours upon selecting Course Code.
- **Cross-Department Course Routing:** Seamlessly routes institution-wide common courses (e.g., General Tamil handled by Tamil Dept faculty, General English by English Dept, French by French Dept, Hindi by Hindi Dept, Sanskrit by Sanskrit Dept, Human Excellence by HE Dept).
- **Strict Even Semester Filtering:** Intelligent semester controls strictly constrained to Even Semesters (II, IV, VI).
- **Standardized Section Selector:** Standardized institutional sectioning (`N/A`, `Shift I - Section A`, `Shift I - Section B`, `Shift II - Section A`, `Shift II - Section B`, `Shift II - Section C`, `Shift II - Section D`).
- **Bloom's Taxonomy CO-K Mapping:** Pre-mapped Course Outcomes (CO1 to CO5) tagged to cognitive domains (K1 to K6).

B. Unit-wise Teaching Plan & Lecture Scheduling (Part-II)

- **Detailed Monthly Teaching Plan:** Flexible multi-unit scheduling permitting multiple units in any single month across the semester timeline.
- **Methodology & Resource Dropdowns:** Standardized drop-down selectors for Teaching Methodologies (Chalk & Talk, PPT Presentation, Flipped Classroom, Group Discussion, Problem Solving, Blended Mode) and Learning Resources (Standard Textbook, Reference Book, Research Papers, NPTEL/SWAYAM, JosTEL e-Content) with dynamic *'Others (Specify)'* input fields.
- **Automated Hour Verification:** Real-time tallying of planned unit lecture hours against prescribed syllabus hours.

C. Active Learning, Assessment & JosTEL Integration (Part-III)

- **Active Learning Activities:** Presets for Quizzes, Seminars, Case Studies, Mind Mapping, and Group Discussions with customizable 'Others' text entry.
- **Continuous Internal Assessment (CIA) Matrix:** Standardized CIA test schema with strict degree-level business logic: Question 3 & 4 fixed at 6 marks for PG and 10 marks for UG; Question 5 (K6 Create level) restricted strictly to PG courses; standardized `DD-MM-YYYY` date formats.
- **JosTEL & ICT Integration:** Configurable Digital Resources and ICT Tools (JosTEL LMS, Google Classroom, Canvas, Edpuzzle, Kahoot) left unpopulated by default for academic autonomy.

D. Post-Submission Faculty Experience & Approval Workflow (Part-IV)

- **Digital Faculty Declaration:** Automated sync between declaration signature and faculty profile.
- **Submission Confirmation Dialog:** Suppresses departmental rosters from faculty view upon submission. Instead displays a dedicated modal: *'Thank you [Faculty Name], your course plan for [Course Code – Title] is recorded and submitted to HoD review.'*
- **Rapid Sequential Submission:** 'Submit for Next Course' button retains faculty and department credentials while resetting course fields for rapid sequential entry.
- **Tab Closure Utility:** 'Close Tab' button for frictionless completion.

E. Role-Based Governance (HOD & Admin Portals)

- **Head of Department (HOD) Portal:** Department-isolated view of all active courses, live submission statuses (Pending, Approved, Revision Requested), full plan review, and one-click official approval.
- **Instant Academic Audit Sync:** HOD approval locks the course plan and automatically commits it to institutional IQAC Academic Audit records.
- **Faculty Roster Management:** HOD privilege to register new or adjunct faculty members directly.
- **IQAC Administrator Dashboard:** Institution-wide macro monitoring across all 34 departments, batch approval oversight, syllabus master catalog management, and export utilities for NAAC Peer Team review.

4. Security, Privacy & NAAC Quality Compliance Matrix

Security / Compliance Dimension	Institutional Implementation Standard	Audit & Accreditation Value
Role-Based Access Control (RBAC)	Three-tier permission hierarchy: Faculty (unauthenticated direct entry), HOD (department password authentication), IQAC Admin (master system authority).	Ensures departmental autonomy while strictly protecting approval authority.
Credential Privacy	Elimination of publicly exposed default passwords on login modals; blank password inputs; secure credential circulation via official confidential PDF circulars.	Prevents unauthorized departmental approvals or tampering with course plans.
Outcome-Based Education (OBE)	System enforces Course Outcome (CO) mapping to Bloom's Revised Taxonomy (K1 Remembering through K6 Creating).	Direct compliance with NAAC Criterion 2.6 (Student Performance & Learning Outcomes).
Pedagogical ICT Integration	Mandatory integration of JosTEL LMS, e-content modules, and blended learning methodologies in teaching plans.	Satisfies NAAC Criterion 2.3 (Teaching-Learning Process & ICT Enabled Tools).
Data Redundancy & Recovery	Dual persistence (MySQL Database + synchronous JSON backup store) + daily automated backups.	Zero data loss tolerance for institutional accreditation documentation.

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